

**INTELLIGENT SYSTEM AND DECISION SUPPORT PRACTICE
EVEN SEMESTER OF 2025/2026 ACADEMIC YEAR**

FINAL PROJECT REPORT



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ENDORSEMENT PAGE

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FOREWORD

I express my gratitude to God Almighty, who continually bestows His grace and guidance, enabling us to complete the Intelligent Systems and Decision Support practicum and the final project report, entitled "Decision Support System for Selecting the Best Airbnb in Paris Using the Streamlit-Based Mamdani Fuzzy Method." This report presents the final project I chose based on the learnings I learned during the practicum.

I would also like to express my gratitude to the practicum assistants who consistently guided and taught me throughout the practicum and in compiling this report. This report is far from perfect, therefore, I welcome constructive criticism and suggestions to improve it.

I would like to express my gratitude to all those who assisted in this writing. I hope this report is useful.

Yogyakarta, 25 Mei 2026

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BAB I

INTRODUCTION

1.1 Background of the problem

The rapid development of information technology has spurred the emergence of various digital platforms in the tourism and accommodation sectors, one of which is Airbnb. This platform offers a wide range of accommodation options with varying prices, locations, facilities, and service quality. The multitude of available alternatives often makes it difficult for users to determine the accommodation that best suits their needs and preferences.

In the process of selecting accommodation, users must consider various criteria such as price, distance to the city center, cleanliness, guest satisfaction, and accommodation capacity. Each criterion has varying levels of importance and is often subjective. Furthermore, the available information is not always certain and tends to be vague (fuzzy), making it difficult for users to make optimal decisions.

These issues demonstrate that the accommodation selection process cannot be carried out simply; it requires a system capable of assisting in systematic and structured decision-making. Therefore, a Decision Support System (DSS) is needed that can process these various criteria and produce appropriate recommendations.

One method that can be used to address this problem is the Mamdani Fuzzy Method. This method can accommodate uncertainty and ambiguity in data by utilizing the concept of fuzzy sets and logic-based rules. With this approach, criterion values can be transformed into linguistic forms such as "low," "medium," and "high," thus making the decision-making process more flexible and closer to human reasoning.

Based on these issues, this research aims to develop a decision support system based on the Mamdani Fuzzy Method that can assist users in determining the best Airbnb accommodation options based on several predetermined criteria.

1.2 Final Project Objectives

The goal of this final project is to design and build a Decision Support System (DSS) that can assist users in determining the best Airbnb accommodation options based on several predetermined criteria.

In more detail, the objectives of this project are as follows:

1. Implement the Mamdani Fuzzy Logistics method in the decision-making process for selecting Airbnb accommodations.
2. Process data based on several criteria, namely price, distance to the city center, cleanliness level, guest satisfaction, and accommodation capacity.
3. Develop an interactive and easy-to-use Streamlit-based application as the system interface.
4. Generate recommendations in the form of rankings of Airbnb accommodations from most recommended to least recommended.
5. Assist users in making decisions more objectively, systematically, and efficiently.

1.3 Benefits of Final Project

The expected benefits from implementing this final project are as follows:

1. For Users

This system can help users choose the Airbnb accommodation that best suits their needs and preferences more quickly, accurately, and efficiently. With this system, users no longer need to manually compare the many available alternatives.

2. For Decision Making

This system provides a more objective and systematic approach to the decision-making process by considering various relevant criteria, resulting in more rational and measurable results.

3. For Scientific Development

This project can serve as an example of the application of the Mamdani Fuzzy Logic method in solving real-world problems, particularly in the field of decision support systems. Furthermore, this project can serve as a reference for further research.

4. For Developers (Students)

This project provides experience in designing and implementing data-driven systems using artificial intelligence methods and enhances understanding of the concept of fuzzy logic and its application in the real world.

5. For Information Systems

This system can be further developed into a more complex application with additional features, such as real-time data integration, additional criteria, and user interface enhancements.

BAB II DISCUSSION

2.1 Basic theory

2.1.1 Decision Support System (DSS)

A Decision Support System (DSS) is a computer-based system used to assist decision-making in semi-structured and unstructured problems. A DSS is designed to process data and information into a more useful form, thus assisting decision-makers in making optimal choices.

In the context of this research, a DSS is used to assist users in selecting the best Airbnb accommodation based on several predetermined criteria, such as price, distance, cleanliness, guest satisfaction, and capacity.

2.1.2 Fuzzy Logic

Fuzzy logic is a method used to handle uncertainty and ambiguity in data. Unlike classical logic, which only recognizes true (1) and false (0), fuzzy logic allows a value to have a degree of membership between 0 and 1.

Using fuzzy logic, a variable can be represented linguistically, such as:

- Rendah
- Sedang
- Tinggi

Pendekatan ini membuat sistem lebih fleksibel dan mampu meniru cara berpikir manusia dalam mengambil keputusan.

2.1.3 Mamdani Fuzzy Method

The Mamdani Fuzzy Method is one of the most commonly used methods in fuzzy inference systems. This method was introduced by Ebrahim Mamdani in 1975 and is widely used in various decision-making problems.

The Mamdani Fuzzy Method has several main stages, namely:

1. Fuzzifikasi

Fuzzification is the process of converting crisp input values into fuzzy values based on a membership function. At this stage, each input value is mapped to a specific membership degree.

Example:

- Price → cheap, medium, expensive
- Distance → near, far

2. Rule Formation (Rule Base)

In this stage, rules are created based on IF-THEN logic that connect input and output variables.

Example:

- IF low price AND short distance THEN high recommendation
- IF high price AND long distance THEN low recommendation

This rule is used as the basis for the inference process.

3. Inference (Implication and Aggregation)

The inference stage is used to determine fuzzy results based on the rules that have been created. This process involves:

- MIN (AND) operation
- MAX (OR) operation

The results of all rules are then combined (aggregated) into a single fuzzy set.

4. Defuzzifikasi

Defuzzification is the process of converting fuzzy results into crisp values. A commonly used method is the centroid method, which calculates the average value of the fuzzy curve area.

The result of defuzzification is a final value used to rank alternatives.

2.1.4 Application of Fuzzy Mamdani in DSS

In this study, the Mamdani Fuzzy Method was used to evaluate and rank alternative Airbnb accommodations based on five main criteria:

1. Harga (Price)
2. Jarak (Distance)
3. Kebersihan (Cleanliness)
4. Kepuasan Tamu (Satisfaction)
5. Kapasitas (Capacity)

Each criterion was processed using membership functions and fuzzy rules to generate a final score. This score was then used to rank the accommodation recommendations from best to least recommended.

2.2 Case Scenario & Dataset

2.2.1 Case Scenario

In today's digital age, booking accommodations can be easily done through various online platforms, one of which is Airbnb. This platform offers a wide selection of accommodations with varying prices, locations, amenities, and levels of service quality. However, this abundance of choices often makes it difficult for users to determine the accommodation that best suits their needs and preferences.

In this case, the user acts as a decision-maker, choosing the best Airbnb accommodation from a range of available alternatives. In this decision-making process, several criteria must be considered, including:

1. Harga (Price)
2. Jarak ke pusat kota (Distance)
3. Tingkat kebersihan (Cleanliness)
4. Tingkat kepuasan tamu (Satisfaction)
5. Kapasitas penginapan (Capacity)

Each criterion has distinct characteristics. Some criteria are cost-based (the smaller the better), such as price and distance, while others are benefit-based (the larger the better), such as cleanliness, satisfaction, and capacity.

The challenge is how to process these various criteria simultaneously to generate optimal accommodation recommendations. Therefore, the Mamdani Fuzzy Logic method is used in a Decision Support System to assist users in determining the best choice based on their preferences.

The ultimate goal of this system is to generate a ranking of Airbnb accommodations from most recommended to least recommended.

2.2.2 Dataset

The dataset used in this project is the Airbnb European dataset obtained from the Kaggle platform. This dataset contains various information related to accommodation listings, such as price, location, capacity, and user ratings.

The data used in this study focuses on the paris_weekdays.csv file, which is part of this dataset. Of all the available attributes, only a few were selected and used as criteria in the decision support system:

- realSum → merepresentasikan harga penginapan
- dist → jarak penginapan dari pusat kota
- cleanliness_rating → tingkat kebersihan
- guest_satisfaction_overall → tingkat kepuasan tamu
- person_capacity → kapasitas maksimum penginapan

Before being used in the calculation process, the data is first normalized to achieve a uniform value range, between 0 and 1. This is done to facilitate the fuzzification process in the Mamdani Fuzzy Method.

This dataset is used as the basis for calculations and generating accommodation recommendations that meet the criteria and weights specified by the user.

2.3 Decision Support System Modeling

In this study, Decision Support System (DSS) modeling was carried out using the Fuzzy Mamdani method by defining input variables, output variables, membership functions, and rules (rule base) used in the decision-making process.

2.3.1 Input and Output Variables

1. Variabel Input

The input variables used in this system consist of five criteria, namely:

1. Price (Harga)
2. Distance (Jarak)
3. Cleanliness (Kebersihan)
4. Satisfaction (Kepuasan)
5. Capacity (Kapasitas)

These five variables are used as the basis for the evaluation process of alternative accommodation.

2. Variabel Output

The output variables in this system are:

- Recommendation Score (Tingkat Rekomendasi)

This output represents the level of suitability of an accommodation and is used to determine the ranking.

2.3.2 Fuzzy Sets and Value Ranges

Before the fuzzy process is carried out, all data is normalized into the range 0 to 1. Each variable has a fuzzy set as follows:

1. Price (Harga) (*Cost*)

- Rendah $\rightarrow [0, 0, 0.5]$
- Tinggi $\rightarrow [0.5, 1, 1]$

2. Distance (Jarak) (*Cost*)

- Dekat $\rightarrow [0, 0, 0.5]$
- Jauh $\rightarrow [0.5, 1, 1]$

3. Cleanliness (Kebersihan) (*Benefit*)

- Rendah $\rightarrow [0, 0, 0.5]$
- Tinggi $\rightarrow [0.5, 1, 1]$

4. Satisfaction (Kepuasan) (*Benefit*)

- Rendah $\rightarrow [0, 0, 0.5]$
- Tinggi $\rightarrow [0.5, 1, 1]$

5. Capacity (Kapasitas) (*Benefit*)

- Kecil $\rightarrow [0, 0, 0.5]$
- Besar $\rightarrow [0.5, 1, 1]$

6. Output (Rekomendasi)

- Rendah $\rightarrow [0, 0, 0.5]$
- Sedang $\rightarrow [0.25, 0.5, 0.75]$
- Tinggi $\rightarrow [0.5, 1, 1]$

2.3.3 Rule Base (Aturan Fuzzy)

Fuzzy rules are used to determine the relationship between input and output variables. Here are some of the rules used in the system:

1. IF Low price AND Close distance THEN High recommendation
2. IF High price AND Far distance THEN Low recommendation
3. IF High satisfaction AND High cleanliness THEN High recommendation
4. IF Low price AND High satisfaction THEN High recommendation
5. IF Close distance AND High satisfaction THEN High recommendation
6. IF Large capacity AND High cleanliness THEN Medium recommendation
7. IF Low price AND Large capacity THEN Medium recommendation
8. IF Far distance AND Low satisfaction THEN Low recommendation
9. IF High price AND Low satisfaction THEN Low recommendation
10. IF Close distance AND High cleanliness THEN High recommendation

2.3.4 Decision Making Process

The decision-making process in this system is carried out through several stages, namely:

1. Fuzzification
Converts input values into degrees of membership in a fuzzy set.
2. Inference (Rule Evaluation)
Uses fuzzy rules to determine output values based on combinations of inputs.
3. Aggregation
Combines the results of all fuzzy rules.
4. Defuzzification
Converts fuzzy results into explicit values using the centroid method.
5. Ranking
The final values obtained are used to rank the accommodation alternatives from best to lowest.

2.4 Calculation & Algorithm

In this study, the method used in the decision-making process is Fuzzy Mamdani. This method was chosen because it can handle uncertainty and produces results that approximate the human way of thinking when evaluating alternatives.

In general, the steps of the Mamdani Fuzzy algorithm used in this system are as follows:

1. Data Collection and Preparation

At this stage, Airbnb accommodation data is collected from a predetermined dataset. The data used covers five main criteria:

- Price (Harga)
- Distance (Jarak)
- Cleanliness (Kebersihan)
- Satisfaction (Kepuasan)
- Capacity (Kapasitas)

Next, the data normalization process is carried out so that all values are in the range 0 to 1, thus facilitating the fuzzy calculation process.

2. Fuzzifikasi

Fuzzification is the process of converting crisp input values into fuzzy values based on a predetermined membership function.

Each criterion value is calculated for its degree of membership within the fuzzy set, as follows:

- Low and High for Price
- Near and Far for Distance
- Low and High for other criteria

The result of this stage is a membership degree value ranging from 0 to 1.

3. Rule Base Formation

In this stage, rules are created based on IF-THEN logic that connect input variables with outputs.

Contoh:

- IF Price rendah AND Distance dekat THEN Rekomendasi tinggi
- IF Price tinggi AND Distance jauh THEN Rekomendasi rendah

This rule base is used as a basis in the inference process.

4. Fuzzy Inference

The inference stage is performed by evaluating all the rules created. This process uses fuzzy logic operators, namely:

- The AND operator uses the MIN function
- The OR operator uses the MAX function

Each rule will produce a fuzzy output, which is then combined (aggregated) into a single fuzzy set.

5. Aggregation

Aggregation is the process of combining the results of all active rules into a single fuzzy output set. This process is performed using the MAX operator to obtain the highest membership value for each rule.

6. Defuzzification

Defuzzification is the process of converting fuzzy results into crisp values. This study uses the centroid method, which calculates the center point of the fuzzy curve area.

The result of this stage is a numeric value that represents the recommendation level of a lodging.

7. Calculating the Final Score with Weights

The defuzzification score is then combined with the weights of each criterion input by the user. These weights reflect the importance of each criterion in the decision-making process. Criteria with higher weights will have a more significant influence on the final result.

8. Ranking

The final stage is sorting all alternatives based on the final score obtained. The alternative with the highest score will be ranked first and considered the best recommendation.

2.5 Implementation & System View

app.py

```
import streamlit as st

st.set_page_config(
    page_title="SPK Airbnb Fuzzy",
    layout="wide"
)

st.markdown("""
<style>
.main {
    background-color: #0e1117;
    color: white;
}

h1, h2, h3 {
    color: #ffffff;
}

.stButton>button {
    background-color: #4CAF50;
    color: white;
    border-radius: 10px;
    height: 3em;
    width: 100%;
}

.stMetric {
    background-color: #1c1f26;
    padding: 15px;
    border-radius: 10px;
}
</style>
""", unsafe_allow_html=True)

st.markdown("""
# Airbnb Decision System
### Fuzzy Mamdani Based Recommendation System
""")

st.markdown("---")

col1, col2 = st.columns([2, 1])

with col1:
    st.markdown("""
### About This Application

This system helps users choose the best Airbnb listing in Europe
using the Fuzzy Mamdani method.

The decision is based on several criteria:

- Price
- Distance to city center
- Cleanliness
- Guest satisfaction

```

```

- Capacity

Use the sidebar menu to explore the dataset and run the decision system.
"""

with col2:
    st.info("Start from the Dataset menu, then continue to SPK Fuzzy")

st.markdown("## System Overview")

col1, col2, col3 = st.columns(3)

with col1:
    st.metric(label="Number of Criteria", value="5")

with col2:
    st.metric(label="Method", value="Fuzzy Mamdani")

with col3:
    st.metric(label="Platform", value="Streamlit")

st.markdown("## Main Features")

col1, col2, col3 = st.columns(3)

with col1:
    st.markdown("""
    ### Dataset
    View Airbnb data directly
    """)

with col2:
    st.markdown("""
    ### Decision System
    Run fuzzy calculation
    """)

with col3:
    st.markdown("""
    ### Ranking
    Display best recommendations
    """)

st.markdown("---")

if st.button("Start"):
    st.success("Please use the sidebar menu")

```

data.py

```

import streamlit as st
import pandas as pd

@st.cache_data
def load_data():
    df = pd.read_csv("dataset/paris_weekdays.csv")
    return df

df = load_data()

criteria_df = df[[
    "realSum",
    "dist",
    "cleanliness_rating",
    "guest_satisfaction_overall",
    "person_capacity"
]]

# Rename biar lebih readable
criteria_df.columns = [
    "Price",
    "Distance",
    "Cleanliness",
    "Satisfaction",
    "Capacity"
]

st.title("Dataset")
st.markdown("---")

#Filter
st.subheader("Filter Data")

col1, col2, col3 = st.columns(3)

with col1:
    max_price = st.slider(
        "Maximum Price",
        int(criteria_df["Price"].min()),
        int(criteria_df["Price"].max()),
        int(criteria_df["Price"].max())
    )

with col2:
    min_satisfaction = st.slider(
        "Minimum Satisfaction",
        int(criteria_df["Satisfaction"].min()),
        int(criteria_df["Satisfaction"].max()),
        int(criteria_df["Satisfaction"].min())
    )

with col3:
    min_capacity = st.slider(
        "Minimum Capacity",
        int(criteria_df["Capacity"].min()),

```

```

        int(criteria_df["Capacity"].max()),
        int(criteria_df["Capacity"].min())
    )

#Apply Filter
filtered_df = criteria_df[
    (criteria_df["Price"] <= max_price) &
    (criteria_df["Satisfaction"] >= min_satisfaction) &
    (criteria_df["Capacity"] >= min_capacity)
]

#Summary
st.markdown("## Data Summary")

col1, col2, col3 = st.columns(3)

with col1:
    st.metric("Total Data", len(filtered_df))

with col2:
    st.metric("Average Price", round(filtered_df["Price"].mean(), 2))

with col3:
    st.metric("Average Satisfaction",
round(filtered_df["Satisfaction"].mean(), 2))

st.markdown("## Data Preview")

st.dataframe(
    filtered_df,
    use_container_width=True
)

st.markdown("## Top 10 (Highest Satisfaction)")

top_data = filtered_df.sort_values(
    by="Satisfaction",
    ascending=False
).head(10)

st.dataframe(top_data, use_container_width=True)

```

fuzzy.py

```

import streamlit as st
import pandas as pd
import numpy as np
import skfuzzy as fuzz
import matplotlib.pyplot as plt

#Load Data
@st.cache_data
def load_data():
    return pd.read_csv("dataset/paris_weekdays.csv")

```

```

df = load_data()

data = df[[
    "realSum",
    "dist",
    "cleanliness_rating",
    "guest_satisfaction_overall",
    "person_capacity"
]].copy()

data.columns = [
    "Price",
    "Distance",
    "Cleanliness",
    "Satisfaction",
    "Capacity"
]

st.title("Fuzzy Mamdani Calculation")
st.markdown("----")

#User Input
st.subheader("User Preference Input")

input_price = st.slider(
    "Preferred Price",
    int(data["Price"].min()),
    int(data["Price"].max()),
    int(data["Price"].mean())
)

input_distance = st.slider(
    "Preferred Distance",
    float(data["Distance"].min()),
    float(data["Distance"].max()),
    float(data["Distance"].mean())
)

input_cleanliness = st.slider(
    "Minimum Cleanliness",
    0, 10, 7
)

input_satisfaction = st.slider(
    "Minimum Satisfaction",
    0, 100, 80
)

input_capacity = st.slider(
    "Preferred Capacity",
    1, 10, 2
)

#Universe
x_price = np.arange(data["Price"].min(), data["Price"].max(),
1)
x_distance = np.arange(data["Distance"].min(),
data["Distance"].max(), 0.1)

```

```

x_clean      = np.arange(0, 11, 1)
x_satisfaction = np.arange(0, 101, 1)
x_capacity    = np.arange(0, 11, 1)
x_score      = np.arange(0, 101, 1)

#MEMBERSHIP FUNCTION

# Price
price_low = fuzz.trimf(x_price, [x_price.min(), x_price.min(),
input_price])
price_med = fuzz.trimf(x_price, [x_price.min(), input_price,
x_price.max()])
price_high = fuzz.trimf(x_price, [input_price, x_price.max(),
x_price.max()])

# Distance
dist_near = fuzz.trimf(x_distance, [x_distance.min(), x_distance.min(),
input_distance])
dist_med = fuzz.trimf(x_distance, [x_distance.min(), input_distance,
x_distance.max()])
dist_far = fuzz.trimf(x_distance, [input_distance, x_distance.max(),
x_distance.max()])

# Cleanliness
mf_clean_low = fuzz.trimf(x_clean, [0, 0,
input_cleanliness])
mf_clean_med = fuzz.trimf(x_clean, [0, input_cleanliness, 10])
mf_clean_high = fuzz.trimf(x_clean, [input_cleanliness, 10, 10])

# Satisfaction
mf_sat_low = fuzz.trimf(x_satisfaction, [0, 0,
input_satisfaction])
mf_sat_med = fuzz.trimf(x_satisfaction, [0, input_satisfaction, 100])
mf_sat_high = fuzz.trimf(x_satisfaction, [input_satisfaction, 100, 100])

# Capacity
mf_cap_low = fuzz.trimf(x_capacity, [0, 0,
input_capacity])
mf_cap_med = fuzz.trimf(x_capacity, [0, input_capacity, 10])
mf_cap_high = fuzz.trimf(x_capacity, [input_capacity, 10, 10])

# Output
out_low = fuzz.trimf(x_score, [0, 0, 50])
out_med = fuzz.trimf(x_score, [25, 50, 75])
out_high = fuzz.trimf(x_score, [50, 100, 100])

#Rule Base
rule_table = pd.DataFrame([
    # HIGH
    {"Rule": "R01", "Kondisi": "Price=Low AND Distance=Near",
"Output": "HIGH"},
    {"Rule": "R02", "Kondisi": "Price=Low AND Distance=Near AND
Cleanliness=High", "Output": "HIGH"},
    {"Rule": "R03", "Kondisi": "Cleanliness=High AND Satisfaction=High",
"Output": "HIGH"},
    {"Rule": "R04", "Kondisi": "Capacity=High AND Satisfaction=High",
"Output": "HIGH"},
    {"Rule": "R05", "Kondisi": "Price=Low AND Satisfaction=High",

```

```

"Output": "HIGH"},
{"Rule": "R06", "Kondisi": "Price=Low AND Cleanliness=High AND
Satisfaction=High", "Output": "HIGH"},
{"Rule": "R07", "Kondisi": "Distance=Near AND Cleanliness=High",
"Output": "HIGH"},
{"Rule": "R08", "Kondisi": "Distance=Near AND Satisfaction=High",
"Output": "HIGH"},
# MEDIUM
{"Rule": "R09", "Kondisi": "Price=Med AND Distance=Med",
"Output": "MEDIUM"},
{"Rule": "R10", "Kondisi": "Cleanliness=Med AND Satisfaction=Med",
"Output": "MEDIUM"},
{"Rule": "R11", "Kondisi": "Capacity=Med AND Satisfaction=Med",
"Output": "MEDIUM"},
{"Rule": "R12", "Kondisi": "Price=Low AND Capacity=Low",
"Output": "MEDIUM"},
{"Rule": "R13", "Kondisi": "Price=Med AND Cleanliness=Med",
"Output": "MEDIUM"},
{"Rule": "R14", "Kondisi": "Price=Med AND Satisfaction=Med",
"Output": "MEDIUM"},
{"Rule": "R15", "Kondisi": "Distance=Med AND Cleanliness=Med",
"Output": "MEDIUM"},
{"Rule": "R16", "Kondisi": "Distance=Near AND Cleanliness=Med",
"Output": "MEDIUM"},
{"Rule": "R17", "Kondisi": "Price=Low AND Distance=Far",
"Output": "MEDIUM"},
{"Rule": "R18", "Kondisi": "Price=High AND Distance=Near AND
Cleanliness=High", "Output": "MEDIUM"},
# LOW
{"Rule": "R19", "Kondisi": "Price=High AND Distance=Far",
"Output": "LOW"},
{"Rule": "R20", "Kondisi": "Cleanliness=Low AND Satisfaction=Low",
"Output": "LOW"},
{"Rule": "R21", "Kondisi": "Price=High AND Satisfaction=Low",
"Output": "LOW"},
{"Rule": "R22", "Kondisi": "Price=High AND Cleanliness=Low",
"Output": "LOW"},
{"Rule": "R23", "Kondisi": "Distance=Far AND Satisfaction=Low",
"Output": "LOW"},
{"Rule": "R24", "Kondisi": "Distance=Far AND Cleanliness=Low",
"Output": "LOW"},
{"Rule": "R25", "Kondisi": "Satisfaction=Low AND Cleanliness=Low",
"Output": "LOW"},
{"Rule": "R26", "Kondisi": "Price=High AND Distance=Far AND
Satisfaction=Low", "Output": "LOW"},
])

with st.expander("Lihat Rule Base (26 Rules)":
    st.dataframe(rule_table, use_container_width=True)

#Kalkulasi
if st.button("Run Calculation"):

    scores = []

    for i in range(len(data)):
        p = data.iloc[i]["Price"]
        d = data.iloc[i]["Distance"]

```

```

c = data.iloc[i]["Cleanliness"]
s = data.iloc[i]["Satisfaction"]
cap = data.iloc[i]["Capacity"]

# — FUZZIFIKASI —————
p_low = fuzz.interp_membership(x_price, price_low, p)
p_med = fuzz.interp_membership(x_price, price_med, p)
p_high = fuzz.interp_membership(x_price, price_high, p)

d_near = fuzz.interp_membership(x_distance, dist_near, d)
d_med = fuzz.interp_membership(x_distance, dist_med, d)
d_far = fuzz.interp_membership(x_distance, dist_far, d)

c_low = fuzz.interp_membership(x_clean, mf_clean_low, c)
c_med = fuzz.interp_membership(x_clean, mf_clean_med, c)
c_high = fuzz.interp_membership(x_clean, mf_clean_high, c)

s_low = fuzz.interp_membership(x_satisfaction, mf_sat_low, s)
s_med = fuzz.interp_membership(x_satisfaction, mf_sat_med, s)
s_high = fuzz.interp_membership(x_satisfaction, mf_sat_high, s)

cap_low_val = fuzz.interp_membership(x_capacity, mf_cap_low,
cap)
cap_med_val = fuzz.interp_membership(x_capacity, mf_cap_med,
cap)
cap_high_val = fuzz.interp_membership(x_capacity, mf_cap_high,
cap)

# — RULE BASE —————

# --- HIGH ---
r01 = np.fmin(p_low, d_near)
r02 = np.fmin(np.fmin(p_low, d_near), c_high)
r03 = np.fmin(c_high, s_high)
r04 = np.fmin(cap_high_val, s_high)
r05 = np.fmin(p_low, s_high)
r06 = np.fmin(np.fmin(p_low, c_high), s_high)
r07 = np.fmin(d_near, c_high)
r08 = np.fmin(d_near, s_high)

# --- MEDIUM ---
r09 = np.fmin(p_med, d_med)
r10 = np.fmin(c_med, s_med)
r11 = np.fmin(cap_med_val, s_med)
r12 = np.fmin(p_low, cap_low_val)
r13 = np.fmin(p_med, c_med)
r14 = np.fmin(p_med, s_med)
r15 = np.fmin(d_med, c_med)
r16 = np.fmin(d_near, c_med)
r17 = np.fmin(p_low, d_far)
r18 = np.fmin(np.fmin(p_high, d_near), c_high)

# --- LOW ---
r19 = np.fmin(p_high, d_far)
r20 = np.fmin(c_low, s_low)
r21 = np.fmin(p_high, s_low)
r22 = np.fmin(p_high, c_low)
r23 = np.fmin(d_far, s_low)

```

```

r24 = np.fmin(d_far, c_low)
r25 = np.fmin(s_low, c_low)
r26 = np.fmin(np.fmin(p_high, d_far), s_low)

# — AGREGASI —————
agg_high = np.fmax(r01, np.fmax(r02, np.fmax(r03, np.fmax(
    r04, np.fmax(r05, np.fmax(r06, np.fmax(r07,
r08)))))))

agg_med = np.fmax(r09, np.fmax(r10, np.fmax(r11, np.fmax(
    r12, np.fmax(r13, np.fmax(r14, np.fmax(r15,
np.fmax(r16, np.fmax(r17, r18))))))))))

agg_low = np.fmax(r19, np.fmax(r20, np.fmax(r21, np.fmax(
    r22, np.fmax(r23, np.fmax(r24, np.fmax(r25,
r26)))))))

# Clipping ke output MF
high_result = np.fmin(float(agg_high), out_high)
med_result = np.fmin(float(agg_med), out_med)
low_result = np.fmin(float(agg_low), out_low)

aggregated = np.fmax(low_result, np.fmax(med_result,
high_result))

# — DEFUZZIFIKASI —————
if np.sum(aggregated) == 0:
    strength_high = float(agg_high)
    strength_med = float(agg_med)
    strength_low = float(agg_low)
    total = strength_high + strength_med + strength_low

    if total == 0:
        final_score = 0.0
    else:
        final_score = (
            strength_low * 25 +
            strength_med * 50 +
            strength_high * 75
        ) / total
else:
    final_score = fuzz.defuzz(x_score, aggregated, 'centroid')

scores.append(round(final_score, 4))

data["Score"] = scores

ranking = data.sort_values(by="Score",
ascending=False).reset_index(drop=True)
ranking.index += 1

st.success("Calculation Completed")

# — STATISTIK SKOR —————
st.subheader("Score Statistics")
col1, col2, col3, col4 = st.columns(4)
col1.metric("Max Score", f"{data['Score'].max():.2f}")
col2.metric("Min Score", f"{data['Score'].min():.2f}")

```

```

col3.metric("Mean Score", f"{data['Score'].mean():.2f}")
col4.metric("Score = 0", int((data['Score'] == 0).sum()))

# — TOP 10 —————
st.subheader("Top 10 Recommendation")
top10 = ranking.head(10)
st.dataframe(top10, use_container_width=True)

# — VISUALISASI BAR —————
st.subheader("Top 10 Score Visualization")
fig, ax = plt.subplots(figsize=(10, 5))
bars = ax.barh(
    [f"#{r} (idx {top10.index[r-1]})" for r in range(1,
len(top10)+1)],
    top10["Score"],
    color="steelblue"
)
ax.bar_label(bars, fmt="%.2f", padding=3)
ax.set_xlabel("Fuzzy Score")
ax.set_title("Top 10 Listings by Fuzzy Mamdani Score")
ax.invert_yaxis()
st.pyplot(fig)

# — DISTRIBUSI SKOR —————
st.subheader("Score Distribution")
fig2, ax2 = plt.subplots(figsize=(10, 4))
ax2.hist(data["Score"], bins=30, color="steelblue",
edgecolor="white")
ax2.set_xlabel("Score")
ax2.set_ylabel("Jumlah Listing")
ax2.set_title("Distribusi Skor Seluruh Listing")
st.pyplot(fig2)

#VISUALISASI MEMBERSHIP FUNCTION
st.markdown("----")
st.subheader("Membership Function Visualization")

tab1, tab2, tab3, tab4, tab5 = st.tabs(
    ["Price", "Distance", "Cleanliness", "Satisfaction", "Capacity"]
)

with tab1:
    fig, ax = plt.subplots(figsize=(10, 4))
    ax.plot(x_price, price_low, label="Low", color="blue")
    ax.plot(x_price, price_med, label="Medium", color="orange")
    ax.plot(x_price, price_high, label="High", color="red")
    ax.axvline(input_price, color="gray", linestyle="--", label=f"Input =
{input_price}")
    ax.set_title("Price MF")
    ax.legend()
    st.pyplot(fig)

with tab2:
    fig, ax = plt.subplots(figsize=(10, 4))
    ax.plot(x_distance, dist_near, label="Near", color="blue")
    ax.plot(x_distance, dist_med, label="Medium", color="orange")
    ax.plot(x_distance, dist_far, label="Far", color="red")
    ax.axvline(input_distance, color="gray", linestyle="--",

```

```

label=f"Input = {input_distance:.2f}"
ax.set_title("Distance MF")
ax.legend()
st.pyplot(fig)

with tab3:
    fig, ax = plt.subplots(figsize=(10, 4))
    ax.plot(x_clean, mf_clean_low, label="Low", color="blue")
    ax.plot(x_clean, mf_clean_med, label="Medium", color="orange")
    ax.plot(x_clean, mf_clean_high, label="High", color="red")
    ax.axvline(input_cleanliness, color="gray", linestyle="--",
label=f"Input = {input_cleanliness}")
    ax.set_title("Cleanliness MF")
    ax.legend()
    st.pyplot(fig)

with tab4:
    fig, ax = plt.subplots(figsize=(10, 4))
    ax.plot(x_satisfaction, mf_sat_low, label="Low", color="blue")
    ax.plot(x_satisfaction, mf_sat_med, label="Medium", color="orange")
    ax.plot(x_satisfaction, mf_sat_high, label="High", color="red")
    ax.axvline(input_satisfaction, color="gray", linestyle="--",
label=f"Input = {input_satisfaction}")
    ax.set_title("Satisfaction MF")
    ax.legend()
    st.pyplot(fig)

with tab5:
    fig, ax = plt.subplots(figsize=(10, 4))
    ax.plot(x_capacity, mf_cap_low, label="Low", color="blue")
    ax.plot(x_capacity, mf_cap_med, label="Medium", color="orange")
    ax.plot(x_capacity, mf_cap_high, label="High", color="red")
    ax.axvline(input_capacity, color="gray", linestyle="--",
label=f"Input = {input_capacity}")
    ax.set_title("Capacity MF")
    ax.legend()
    st.pyplot(fig)

```

profile.py

```

import streamlit as st

st.title("Profile")
st.markdown("---")

st.subheader("Project Information")

st.write("""
**Sistem Pendukung Keputusan (SPK)** untuk menentukan rekomendasi
penginapan Airbnb terbaik menggunakan metode **Fuzzy Mamdani**.

Sistem ini mempertimbangkan beberapa kriteria:
- Price (Harga)
- Distance (Jarak)
- Cleanliness (Kebersihan)

```

```

- Satisfaction (Kepuasan)
- Capacity (Kapasitas)
"""

st.subheader("Team Members")

col1, col2 = st.columns(2)

with col1:
    st.write("**Nama:** Marvel Valensiano")
    st.write("**NIM:** 123240019")

with col2:
    st.write("**Nama:** Rio Prasetyo")
    st.write("**NIM:** 123240136")

st.subheader("Course Information")

st.write("""
Mata Kuliah: Sistem Pendukung Keputusan
Metode: Fuzzy Mamdani
Tools: Python, Streamlit, Scikit-Fuzzy
Dataset: Airbnb Europe Dataset
""")

st.markdown("---")
st.write("© 2026 - Final Project SCPK")

```

Image 1. Home Page

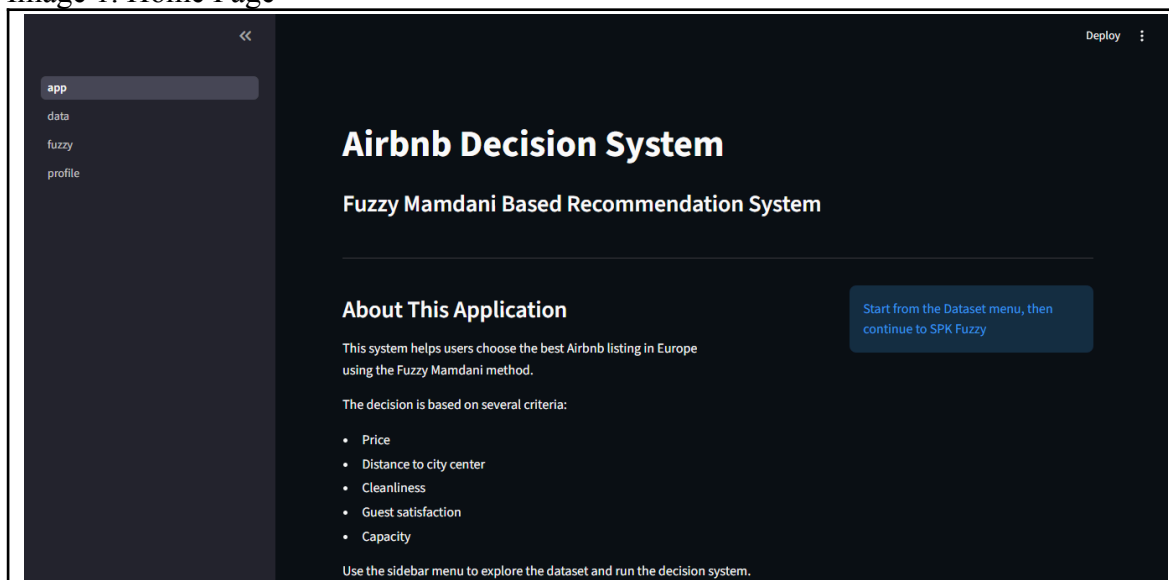


Image 2. Data Page

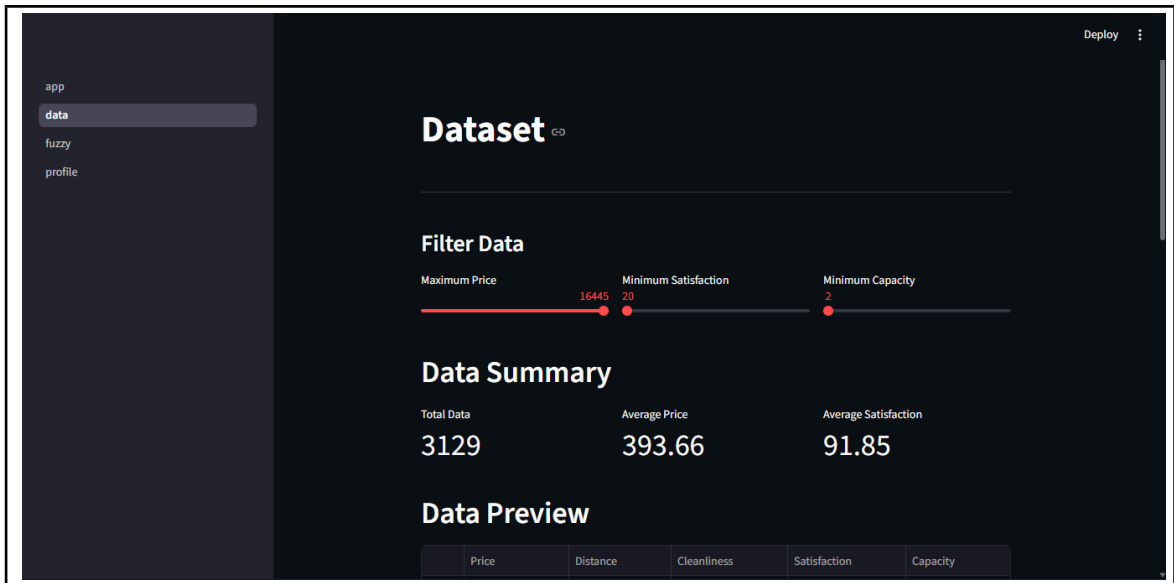


Image 3. Fuzzy Calculation Page

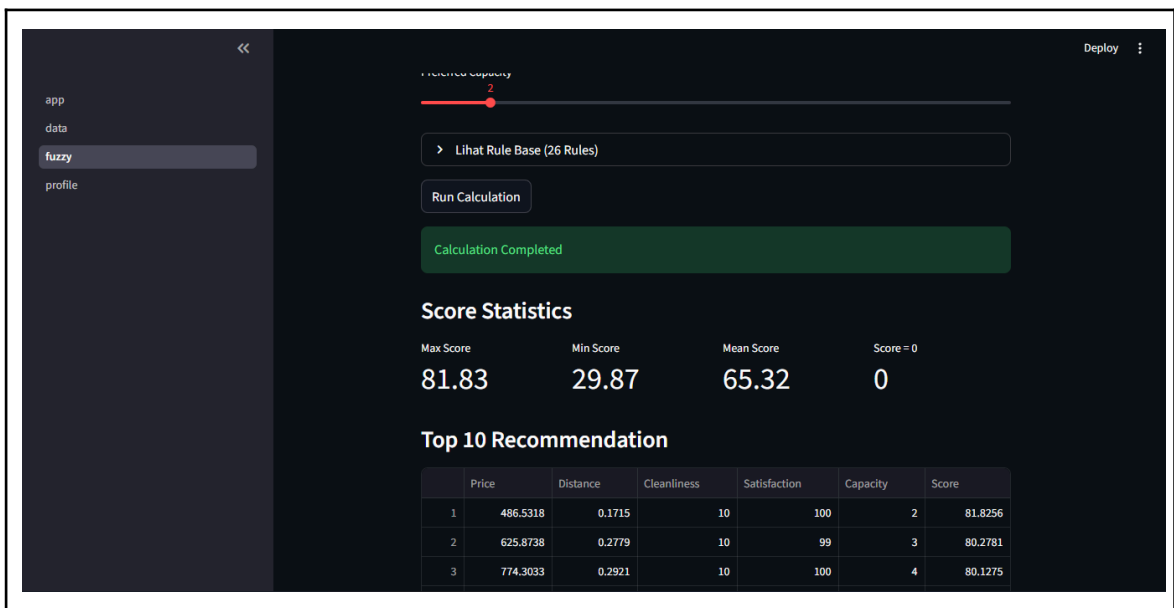
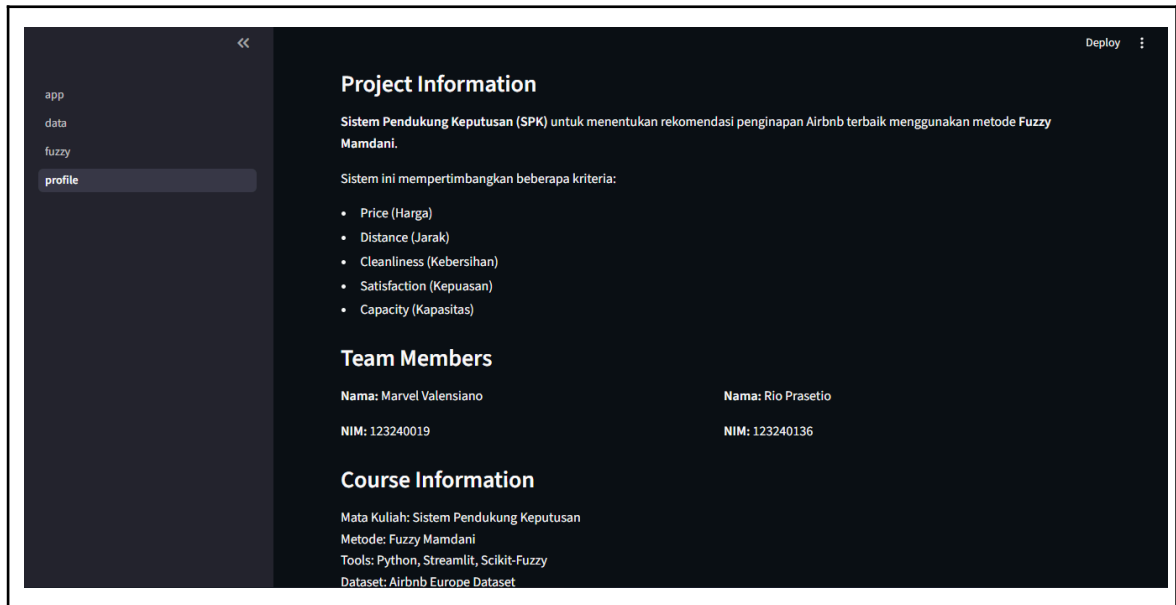


Image 4. Profile Page



BAB III WORK SCHEDULE AND TASK DIVISION

3.1 Work Schedule

No	Activity	2026			
		Week			
		1	2	3	4
1	Literature Study	✓			
2	Dataset Collection	✓	✓		
3	System Requirement Analysis		✓		
4	System Design		✓	✓	
5	Implementation(Coding)			✓	✓
6	System Testing				✓
7	Report Writing	✓	✓	✓	✓
dst					

3.2 Task Distribution

No	Name	NIM	Task Description
1	Marvel Valensiano	123240019	- Writing the report (Chapter I – IV) - Preparing project documentation - Case study and dataset analysis
2	Rio Prasetyo	123240136	- Developing the application (Streamlit) - Implementing Fuzzy Mamdani method - Data processing and visualization

BAB IV

CONCLUSION AND SUGGESTIONS

4.1 Conclusion

Based on the design, implementation, and testing of the Decision Support System (DSS) using the Fuzzy Mamdani method, it can be concluded that:

1. The system has been successfully developed to assist users in selecting the best Airbnb accommodation based on multiple criteria, namely price, distance, cleanliness, satisfaction, and capacity.
2. The implementation of the Fuzzy Mamdani method is able to handle uncertainty and provide a more flexible decision-making process through the use of fuzzy logic and rule-based evaluation.
3. The system produces a final recommendation in the form of a ranking of alternatives, starting from the highest score (most recommended) to the lowest score.
4. The addition of dynamic weighting allows users to adjust the importance of each criterion according to their preferences, making the system more adaptive and user-oriented.
5. Overall, the developed system is capable of providing decision support in a more systematic, objective, and efficient manner compared to manual selection.

4.2 Suggestions

For future development, several improvements can be considered:

1. The system can be enhanced by adding more criteria, such as facilities, location type, or user reviews, to improve the accuracy of recommendations.
2. The rule base in the Fuzzy Mamdani method can be expanded to cover more complex decision scenarios and improve the quality of the results.
3. The system can be integrated with real-time data sources or APIs to provide up-to-date information.
4. The user interface can be further improved to enhance usability and provide a better user experience.
5. Future research may explore the use of other decision-making methods or hybrid approaches to compare performance and improve system effectiveness.

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